

ActInf Livestream #035.2 ~ "A tale of two architectures free energy, its mod

Livestream | 1:50:18

Hello and welcome everyone.

It is January 12th, 2022.

We are here in ActInfLab livestream number 35.2, discussing the paper A Tale of Two Architectures, Free Energy, Its Models, and Modularity.

Welcome to the ActInfLab.

We are an online participatory lab that is communicating, learning, and practicing applied active inference.

You can find more information on this slide.

This is recorded in an archived livestream, so please provide us with feedback so that we can improve our work.

If you are watching live, feel free to write questions in the live chat.

If you are watching in replay, you can also add comments.

All backgrounds and perspectives are welcome here.

And we will be following good video etiquette for livestreams.

All of our activities in the lab are participatory.

So if you would like to contribute or get involved, go to activeinference.org and see what is there.

Today in livestream 35.2, we're going to be having our third discussion continuing on the paper A Tale of Two Architectures, Free Energy, Its Models, and Modularity,

written by Majid Beni, who we're really appreciative is joining us today to talk and see where it goes.

So we'll get to the introduction in a second.

But basically, we're going to just pick up on some of the threads that we opened in earlier discussions.

We have a few specific things written down, and I'm sure we'll go many different ways with people joining and ideas coming up and questions in the live chat and all of that.

So we can kind of go any which way, but we'll just start with the introduction and warm-up.

So we can each say hello.

Maybe Blue and I can just say what we're excited about for the .2.

And then Majid, it'd be awesome to hear any context for the paper.

So I'm Daniel.

I'm a researcher in California.

And I didn't expect the rabbit duck to appear in .1.

But I'm looking forward to continuing that and seeing how is this drawing like a scientific model.

And then what does that philosophical or conceptual paradox or ambiguity say about scientific modeling?

So Blue?

I'm Blue.

I'm a researcher in New Mexico.

And I am excited to probe deeper into the relationship between agents and their environment and also the intransitivity argument and modularity and how those kind of all fit together.

Excellent.

Majid?

You are muted.

And then please continue.

So hello.

You can't hear me?

No?

Yes.

So I'm Majid.

I'm an assistant professor of philosophy at Mechu Middle East Technical University.

And I'm obsessed with models and structures across science and perception and mind.

And it's for a while that I'm engaged with this issue of applying model-based approach to science, to free energy principle and predictive coding.

And I'm very happy to be here.

Thank you very much for inviting me.

Great.

Well, maybe just for like background and context.

What brought you to write this specific paper?

Like recognizing modularity as an important topic?

And where does this fit into some of your other threads related to FEP and model-based science?

Well, that's a very good question.

I'm not quite sure why.

So one reason for focusing on modularity in this paper is that most probably it has some sort of psychological explanation because I started to learn about philosophy of mind and philosophy of cognitive science by reading this language of thought.

And I think that even in the earlier stages when people were studying there, no one was thinking that the brain is actually a syntactical engine.

No one was thinking that if you open up a brain or a cognitive system, you actually find things that are inside the brain in terms of propositional-like entities, language-like entities or syntactical engines or anything of that sort.

But at the same time, people were believing that that perspective on how the brain works has some sort of explanatory gain.

There is something that we can learn by taking that perspective on the brain and cognitive system.

And well, I think that the general idea behind applying this model-based approach to sciences is that it is the explanatory interest, explanatory as well as practical interest of the modeler, of the people who are engaged with scientific enterprise that provides a handle,

on the target system, the things that we try to explore and understanding.

And well, I think that when I was reading the target paper by Kirchhoff and Hippolytov, what struck me as odd was that they were taking it very, very much for granted through, of course, through the very well-articulated scientific and philosophical arguments that there is no such modularity inside the brain or cognitive system.

And I just wanted to bring the point to attention that it would be a matter of, as I said earlier, practical and explanatory goals and interests of the modeler, whether there is modularity there or not.

And that's what I'm going to emphasize, underline in this session, is that this model relativity doesn't need to lead to downright anti-realism about theories of brain and cognitive system.

And that's what I'm going to emphasize.

And I think that model relativity is not all about, you know, the model relativity is not all about taking an anti-realist attitude toward brain science.

But at the same time, so the old goal, the old project is to end up with some sort of subtler perspective on scientific realism about cognitive science and brain science.

[illegible]

Thanks for being here, Majido.

Really glad that you're here to talk about this paper.

I found the paper fantastic.

Pleasure is all mine.

So, modularity.

I mean, it's in the title.

Modularity, models, and then the free energy,
and then the tale of two architectures,
which was a little bit of a reference towards the tale of two densities,
another active paper.

So, just how would you reframe the argument of Hippolito and Kirchhoff Majid?

And where do you think, like, in 22,
what can we say now that we couldn't have said in 2019?

Well, I do not think that we could say anything more than we were saying in 2019, because all of the actual stuff was there.

I think that this is something that comes with philosophy, because no information has been added.

As a matter of fact, I just elaborated on the information that was there in terms of the argument from intransitivity and all of the discussion of causal modeling and dynamical modeling and all that.

So, I don't think that any actual information has been added.

The only thing that has been elaborated on was the emphasis on the rule of the modeler.

The, which, if we insist on,

so, as I was watching this in the previous discussion of,

of the same paper in Active Inference Lab,

and I was, so, Dean's discussion of the,

in terms of the Bob-Alice metaphor,

the discussion of the perspective of the observer,

the observer that is watching both Bob and Alice,

interested me very much.

So, if I want to rephrase the discussion of the modeler

in terms of information theory,

we can as well start restate that in terms of the observer.

But I think, I don't think that the point is that technical.

The point is rather philosophical than scientific or formal

in terms of information theory.

So, no actual information has been added during these two years.

Only, and I think that this approach is gaining momentum.

This emphasize on the rule of modeler or interpreter

is drawing even more,

is leading to even more discussions

in the pre-energy principle literature.

I totally agree with that point.

And then, Lourdes Dean,

it made me think of quantum

and how the double-slit experiment,

kind of kicking and screaming,

has taken a hundred years

in that one limited setting

to consider the role of the observer,

even though it's a really complex thing.

We won't go into it,

but we'll talk about quantum and FEP in a few weeks.

But that was like the anomaly

that signaled,

yeah, you need to take the observer into account

as part of the measuring apparatus.

Very quantitative, empirical observation

that there was a non-negotiable
or irreducible role of the observer.
And then on the qualitative side,
the relational insight
that scientists are human
and science is a human activity.
And that was not driven
on empirical observations
of wave and particle on a screen,
like the twin slit was,
but it was just a sort of
low-level philosophical argument
that there isn't a fungible role
for the observer or the modeler.
And so it is very interesting
to see how that has started to catch on.
And then how does that change
the way we do science
when that is taken
as like a starting point
that there's an observer
modeling the system?
So like that was in this diagram
when we started with
just talking about like
the functional and the effective connectivity
and then the mechanism,
which is kind of not either of those.
And then it wasn't until the very end
when we're like,
oh, but it's the scientist
projecting in
to this whole model
that's another level
of how we think about it.
We can't just like
all be sitting around
just talking about Bob and Alice,

but then now there's going to be us
with multiple perspectives
looking at the scientists
taking multiple perspectives
on Bob and Alice.

So are we stuck in this sort of
subjective infinite regress?

How are we going to salvage
cognitive realism?

Are you asking me?

Anyone.

But if you have any thoughts,
definitely go for it.

I have some ideas,

but I mean,

I prefer other...

How about, yeah,

Dean and then Majid

and then Blue if you want to.

Well,

one of the things, Majid,

that I took away

from reading your paper

was...

This is funny

because typically

when I pick something up,

it's pretty foreign to me.

It starts out being

way on the other side

of a Markov blanket.

And then I guess I somehow

I'm able to sort of

get inside of it

and relate it back

to the things that I know.

and eventually I get

a bit of a grip on it.

And then next thing you know,
there's four duck rabbits
spinning madly out of control
on a gradient.

But one of the things
that I took away from this
was...

And I think I mentioned it
in the point one is...
Is...

Are you nine o'clocking?
When we're taking these slices
in time

and we're building
these models,
my question was,
what are the specifics of that?
I mean,
if we take an FMRI image,
I know that the machine itself
that's taking those slices of time
isn't necessarily conscious,
but it is carrying out
some sort of an instantiation exercise.

And so one of the things
I wanted to kind of talk
to you a little bit about,
not necessarily on a quantum level,
but one of the sections
of your paper,
which was really,
really influential to me,
was a section on a golden thread.

I wanted to...

I was hoping that perhaps
within that idea
of partitioning,
whether it's through

quantum methodology
or quantum series
or whatever,
if you could maybe
talk a little bit
about that idea
of what you meant
by a golden thread,
not just metaphorically,
but maybe materially
and maybe influentially,
if there is such a word,
because that part of it
really mattered to me
and it actually got me
writing a whole set
of subnotes
based on that section.
So maybe you could expand
on that a little bit
because I think
there's a lot there.
Thanks, Dean,
for the great question.
So yes,
we'd love to learn more
about the golden thread.
Unmute and then continue.
I told that...
So very well.
So if I may do
a bit of advertisement,
I have been discussing
this specific issue,
this dog-rabbit thing,
in another paper,
which is published
in a philosophy journal.

It is named

Cognitive Penetration

and Cognitive Realism,

and it is published

in Episemm.

It relates this issue

to the issue

of theory-ladenness

in philosophy of science,

works of Thomas Kuhn

and all that.

So I can't...

So I can't...

Well, what I say here...

So I'm also interested

in the relation

between FIP and quantum theory.

And I'm working on a project

that relates FIP

to cubism

as one of the significant

interpretations

of quantum mechanics.

But in this context,

I cannot...

So the paper

is not directly related

or even indirectly

to quantum mechanics

and the rule of observer.

I rather take

some sort of...

So if I want to state it

in more technical terms,

I would use

Shannon's theory

of information

to articulate

the rule of the observer
a bit clearer.

So...

About the question,
if I understand it correctly,
what happens
when the picture rotates
and there's a scanning
or some sort of...
scanning is taking place?
Whether...

So...

Am I correct to understand
that your question
is about what happens
when there is a rotation
but there is no observer
to want to work out
whether it is a duck
or a rabbit, right?

Is it...

Yeah.

And the whole business
of so...

So...

When we're trying
to encapsulate something
and you spoke
clearly in the paper
about...

We don't necessarily
have to see this
as a hard and fast
edges kind of thing.

We can kind of appreciate
that there's a certain
amount of fuzziness
that's actually enabling

our conversation
and so that...
Because the lines
aren't abrupt
and when we do decide
to put a Markov blanket
around something
it's us choosing
to do that
at a certain scale
with a certain amount
of specificity
and Stephen Sillit
in the point one
talked about
so what...
What is available
at a meso scale?
So that golden thread
business
that...
What do we lasso?
What is that fMRI
actually trying to entail?
That's what I was hoping
that you could maybe
tease out for us
a little bit.
Very well.
So I can't be all clear
about that
because being a philosopher
I'm a bit naive
about neuroimaging
and that sort of stuff
but as some sort of
general insight
and trivial insight

from philosophy of science

I would say that

it would depend on

what we are trying

to trace

by using fMRI

or any other

observational device

or measurement

so...

And...

The fact that we could use

these measurement devices

in a number of different ways

and by devices

I do not only refer

to actual devices

but also theoretical devices

such as Markov blankets

and that sort of stuff

so...

And applying that

to different scales

and different levels

doesn't need to mean

that there is no

objective fact of matter

to be represented

by these devices

so...

I might be able

to use different

theoretical and

mathematical devices

to measure

the...

space of my room

I may be able

to use
for example
kilometers
or I may be
able to use
width nodules
or something of that sort
and...

This doesn't mean
that there is no
fact of matter
to be measured
after all
we can see this picture
in terms of
either
rabbit
or

a duck
but not as a dog
or a cow
for example
and whether we see it
as a duck
or a rabbit
again
is influenced
by...

So...
I would say that
to some extent
it will depend
on the original
vagueness
of the picture
to what extent
the subjectivity
to what extent

can affect
the perception
what we see
is a matter of
the target system
itself
the properties
of the target
system itself
so...
I mean
no matter
how hard we try
we could hardly
be able to see
this duck or rabbit
as a cow
or a dog
or something else
but at the same time
to some extent
especially when
the target system
allows for
some vagueness
to some extent
or presuppositions
or goals
and intentions
be around
the actual situation
and...
So there is
some sort of
constraint
or
moderate amount
of

theory-relatedness
to use a jargon
by philosophers
of science
that is taking place
here
but
it doesn't mean
that there is
no...
I mean
it doesn't need
to strip
the target
of all of its
objective features
I still...
I'm still not sure
whether I could
address your worry
correctly or not
because I couldn't
see the point
about the
fMRI
and neuroimaging
techniques
so...
Well...
I was...
I guess I'm kind of...
I'm agreeing
with the argument
that you were
putting forward
which is that
any time
we make an observation

and then we
draw up some kind
of a representation
we call that a model
even in those...
even in those efforts
to try to be precise
we have to leave
open a little bit
of latitude
a little bit
of wiggle room
because
as you pointed out
you can have a
dynamic causal model
and get into
an argument
whether or not
that's the appropriate
model that should be
an acyclic one
right?

For example
that was what
you were saying
and
that kind of
gets you
going down
a path
of arguing
about
which is the
right tool
as opposed
to saying
do tools

leverage
right?
I think that's
what...
maybe that's
what I was reading
into
what you were
trying to say
that's all
thanks Dean
Blue
and then
Majid
so I think
this kind of
goes into
what Dean
was saying
about
scale-friendly
versus
scale-free
and so
like with
a Markov
blanket
and with
like this
you know
denotation
of a module
whether it's
vertical
or horizontal
or vertical
module
you know

we impute
the Markov
blanket
on whatever
scale
we're trying
to separate
conditionally
so I just
wanted to
bring that
point
back in
and also
this idea
of maybe
like
incomplete
partitioning
so I mean
if there was
a complete
partition
around each
module
there would
be like
no ability
to transfer
information
from module
to module
and in the
brain
I mean
we do
know
that definitely

some
the pieces
have to
communicate
to make
the entire
brain
or it
wouldn't be
considered
an organ
you know
we wouldn't
be able
to Markov
blanket
you know
this idea
of nested
Markov
blanket
so I
don't know
about this
information
encapsulation
or information
leakiness
Majid
it would be
nice if
you maybe
kind of
unpacked
that a
little bit
more
for us

yes the
information
encapsulation
topic
so the
thing is
that I
can't see
for some
reason I
can't see
you in
this picture
and I
have to
shift to
the YouTube
page in
order to
see you
you can
maybe go
to on the
bottom bar
toggle tile
view
I use
that
it's
for some
reason
I can
see that
there are
pages of
people but
I cannot
just see

your faces
it shows
the initial
D and
D and
B
it may
have to
do with
what countries
we're all
in which
is totally
fine
at least
we have
audio
let me
try it
once more
well
yes at
least we
have audio
so very
well
well
about
information
encapsulation
shall I
try to
explain that
in terms
of that
Bob
Alice
metaphor

great
yeah
information
encapsulation
as applied
to the
Bob
Alice
parable
all right
very well
so
for one
thing
this
information
encapsulation
is one
of the
features
of
Jerry
Fodor's
original
statement
of
the
thesis
of
modularity
and
so
which
is quite
flexible
a flexible
thesis
and

models
so
for me
in the
way
that
I see
that
model
relativity
entails
this kind
of
modularity
information
encapsulation
because
in order
to have
model
and
again
it can
be
stated
in terms
of
Markov
blankets
and
whether
the
Markov
blankets
are
embodied
or
nested

or
not
is
itself
another
issue
but
the
very
fact
that
we
can
use
Markov
blankets
or
Markovian
models
it indicates
that we
can
we are
able to
set
some
sort
of
barriers
conditional
independence
between
internal
and external
spaces
so
in order
to state

this free
energy
principle
person
interpret
these
spaces
in terms
of
active
spaces
and
sensory
spaces
but
at the
same
time
it
indicates
that
there
is
some
sort
of
barrier
even
though
if
the
barrier
is
conditional
and
probabilistic
I
do

not
think
that
for
information
encapsulation
we need
anything
stronger
than
this
kind
of
flexible
conditional
barrier
between
inside
and
outside
or
between
different
spaces
and
this
is
the
point
that
so
when I
was
watching
your
discussion
in
the

previous
session
035.1
I
saw
that
this
metaphor
of
Bob
and
Alice
came out
and
there
was
the
question
of
whether
there
is
any
sort
of
!
relation
between
them
and
whether
the
relation
between
them
is
any
way

correlated
so
as I
argued
in
the
paper
there
may
be
some
sort
of
causal
relation
or
correlation
which
can
be
stated
in
terms
of
interventionist
approaches
which
are
liable
to
other
sources
of
criticism
that
we
don't
need

to
discuss
right
though
but
I
think
that
information
encapsulation
doesn't
need
anything
stronger
than
the
general
model
relativity
that
I
tried
to
advocate
in
this
paper
so
the
main
point
is
that
regardless
of
whether
or not
there

is
any
kind
of
causal
relation
actual
relationship
between
the
actions
of Bob
and
the
actions
of
Alice
or
perhaps

I
have
to
call
them
Charles
and
Lucy
in
order
to
be
more
loyal
to
the
soul
of

the
tale
of
two
cities
the
thing
is
that
to
the
extent
that
the
information
in
each
unit
is
not
available
accessible
to
the
other
unit
we
can
speak
of
information
encapsulation
and
when
we
are
making
models

if
we
can
set
any
kind
of
distinction
between
internal
and
external
spaces
and
if
we
can
in
the
same
way
that
we
may
not
be
able
to
set
a
distinction
between
models
and
their
target
there
wouldn't

be
any
modeling
practice
there
wouldn't
be
any
scientific
activity
on
the
other
hand
if
we
are
able
to
set
a
distinction
between
internal
and
external
or
any
kind
of
distinction
between
models
and
their
we
would
be

able
to
state
some
sort
of
conditional
barrier
it
still
doesn't
indicate
that
there
is
no
kind
of
information
so
setting
the
barrier
by itself
doesn't
show
that
the
information
in
one
block
is
not
available
to
the
other

block
but
I
can
see
the
yeah
I
was
fascinated
by
this
picture
when
I
was
watching
your
discussion
in
the
previous
session
so
in
terms
of
the
relationship
between
Bob
and
Alice
the
action
of
brushing
the

teeth
is
correlated
with
what
Alice
does
in
the
other
unit
but
it
has
been
assumed
that
the
information
in
the
unit
is
not
available
the
information
in
Bob
unit
is
not
available
to
Alice
and
vice
versa

even
though
the
information
may
be
available
to
us
as
observers

or
if
the
!
other
observer
a third
person
the
information
might
be
available
to her
to the
observer
but
to the
extent
that the
information
in each
unit
is not
available
to the

person
in the
other
units
I think
that we
can speak
of some
sort of
partial
information
encapsulation
thanks
very
interesting
if we
couldn't
I just
really
caught in
that part
about how
if we
couldn't
make a
distinction
between
the
target
system
and the
model
between
the
territory
and the
map
then it's

not
modeling
so it's
kind of
like well
when you
kick
a football
does it
model
a
parabola
well it
doesn't
have
the
modeling
the
self
modeled
integrated
world
model
whatever
one wants
to call
it
such that
it is
just
the
actual
causal
relationships
that it's
engaging in
the world
and no

more
whereas
cognitive
entities
like
scientists
are
engaging
in
modeling
and then
that means
that they
are able
to engage
in
counterfactuals
in thinking
through other
minds
they are
reducing
their
uncertainty
from
a
partial
information
encapsulation
view
and sometimes
you get
more
information
than the
system
has
available

but other
times it's
radically
less
and so it's
kind of like
an application
of a
Markov
blanket
partitioning
with some
non-Markov
concepts
from philosophy
of science
to the
scientist
investigating
target
systems
getting
partial
measurements
that's
their
observations
coming in
and then
they're
updating
their
hidden
state
inference
about what
they think
is happening

the likelihood
of different
hypotheses
just like
any active
entity
would
well if I
may elaborate a
bit more on
this and it
returns to the
question of
cognitive
realism
I think that
despite speaking
of all of this
model
relativity
doesn't need to
lead to some
sort of global
or general
form of
anti-realism
because
well a part of
my activity
in the past
years was
dedicated to
developing some
sorts of
cognitive realism
that is based
on FET
assuming that

the practice
of scientific
model making
is underpinned
by the
dynamical
interaction
between
scientists
or groups
of scientists
and their
environment
so this
kind of
model
relativity
I mean
the relation
between
free energy
principle
and
model-based
science
is very
complicated
and it is
not something
that is directly
related to
this aspect
of them
of this
paper
but this
paper is
a piece

of a
larger
project
that has
been
finding its
own shape
in the past
few years
and it is
about the
as I say
the general
idea here
is that
in the
same
even though
the
free energy
principle
the theoretical
framework
experimental
activity
that
is
embodied
in the
research
stream
that leads
to the
development
of free
energy
principle
is

basically
some sort
of
model-based
scientific
activity
on the
other
hand
if we
are
interested
in
explaining
scientific
activity
and
scientific
model
making
and if
we are
naturalist
we are
interested
in finding
good
natural
explanations
for it
we have to
look into
the underpinning
natural
mechanism
that makes
scientific
model

making
possible
right
and a good
place
to look
for that
is
the
scientific
accounts
of the
interaction
between
the agent
the organism
and the
environment
and because
free energy
principle
provides a
viable
account
of the
relation
between
agents
or as
I discuss
in my
more recent
work
a group
of
agents
which is
the

community
of
scientists
and their
environment
both
social
environment
and biological
environment
we can
as well
grow
scientific
model
making
in the
mechanisms
of
free
energy
principles
so
the
situation
is
quite
complicated
and
runs
in both
direction
we like
running
in both
directions
so I
guess

that's
okay
so we
have a
few
different
ways we
could go
Dean or
Blue do
you want
to ask
a question
or we
can go
to one
that we've
previously
written
down
can I
ask a
quick
question
Daniel
yes
yeah
so
in the
in the
point one
Blue brought
up something
that I'm
still sort
of
trying to
untangle

when she
talked
she was
asking
about
genes
in
particular
and how
it might
relate
to what
this paper
was
potentially
pointing
to
at least
it raised
the specter
around
the idea
of
so
information
encapsulation
do we
need
time
in order
for us
to be
able
to
take
a
measure
or

put
a
boundary
around
or
encapsulate
or
somehow
mark
in a
Markovian
way
decide
where
the
partition
is
and
so
I'm
still
trying to
figure out
whether
or not
we are
talking
about
a world
where
because I
think a
world exists
where there
are two
things
going
into

not
necessarily
the
same
direction
we got
one thing
where we
look at
the
world
and we
ask
ourselves
whether
something
was
completed
we do
a
completion
analysis
we take
an
account
and I
think
models
fit
nicely
into
that
world
what I
thought
this
paper
raised

some
ideas
around
was
is
that
half
of
the
whole
picture
is
the
other
half
do
models
and
being
modular
allow
us
to
include
or
make
sure
that
we
have
included
everything
that
we
need
in
order
to

be
able
to
get
the
best
sense
of
what's
on
the
other
side
of
that
divide
so
when
we're
making
these
inferences
we
can
use
models
as
a
way
to
account
for
things
but
are
we
also
able

to
project
off
of
that
and
sort
of
make
better
guesses
better
inferences
I
use
the
expression
prediction
matter
expert
and
so
I
was
wondering
again
those
words
weren't
exactly
used
in
this
paper
but
was
that
part

of
what
we
were
trying
to
sort
of
dig
down
to
is
it
is
it
just
a
matter
of
people
use
models
people
people's
models
account
for
things
in
certain
ways
but
they
also
if
we
don't
get

too
hung
up
on
the
precision
of
that
account
they
can
also
be
sort
of
fuzzy
and
left
semi
open
and
give
us
an
opportunity
to
also
be
creative
and
decide
essentially
whether
or not
we
have
enough
information

to
make
that
next
best
guess
so
just
kind
of
wondering
about
that
may
I
yes
please
all right
very well
that's
a very
brilliant
point
so
it
comes
back
to
the
issue
of
completeness
of
models
their
accuracy
certainty
and that

sort
of
stuff
and
completely
accurate
so
it
has
been
taken
for
granted
and
I
think
that
part
of
this
interest
has
been
raised
by
Richard
Levin's
paper
the
strategy
of
model
based
science
in
population
biology
that

we
are
making
models
but
the
accuracy
of
models
is
very
important
so
the
models
have
to
be
accurate
enough
to
represent
to
do
their
representational
task
well
and
they
have
to
be
complete
in
the
sense
that

they
have
to
be
detailed
enough
I
think
that
this
is
very
good
point
but
look
how
much
accuracy
is
enough
I
mean
we
can
speak
of
completeness
of
models
but
absolutely
complete
models
I
think
that
I've

been
reading
a
short
story
from
American
writer
on
the
exactitude
of
science
or
something
like
that
which
is
the
story
of
people
in
an
empire
who
are
trying
to
make
maps
of
the
empire
and
they
are

very
obsessive
they
want
to
have
complete
maps
but
they
end
up
with
having
maps
that
are
oh
all right
perfect
yes
yes
precisely
so
they
end
up
with
having
maps
that
are
as
the
empire
itself
so
it's

completely
true
that
Levins
2
speaks
of
accuracy
and
detailedness
of
models
but
then
again
completeness
in
it
so
details
I mean
more details
are not
necessarily
leading to
better
models
because
accuracy
works
only in
combination
with
some
other
virtues
such
as

generality
and
precision
of
prediction
so
even
though
so
the
other
question
is
that
how
to
find
how
to
strike
the
right
balance
between
generality
on one
hand
and
accuracy
and
precision
of
models
on the
other
hand
and
I'm

tempted
to
say
that
I
understand
that
I'm
just
trying
to
point
to
the
same
direction
to
speaking
of
the
rule
of
models
scientific
models
and
model
making
but
then
again
how
accuracy
have
to
be
taken
into

account
and
have
to
be
related
to
have
to
put
in
harmony
with
generality
again
I would
I'm
tempted
to say
that
it is
again
a matter
of
interest
and
explanatory
and
predictive
goals
of the
model
maker
so
I will
ask
a
question

from
the
live
chat
from
Stephen
so
Stephen
Majid
was
curious
about
your
understanding
of
the
relationship
in which
practitioners
seek to
apply
low
dimensional
computational
modeling
with other
scales
of
knowing
in
uncertain
and
dynamic
environments
so
for
example
there's

the
awareness
and the
perception
of
the
duck
rabbit
and
then
there
are
the
affordances
in
our
niche
that
go
beyond
just
sort
of
a
visual
toy
example
so
open-ended
question
about
what
you might
think
about
that
kind
of

application
of
modeling
perhaps
in a
social
setting
do
do
the
models
in
the
social
setting
need
to
be
dimensional
or
what if
I see
that
in terms
of
the
relation
between
dynamical
causal
models
and
DAGs
so
for
example
I
assume

that
DAGs
could provide
some
sort
of
low
dimensional
model
that
is
capable
of
representing
the
network
aspects
or
something
like
that
yeah
I
think
that's
a
good
question
I
mean
continue
with
that
and
I'll
write
some
things

down
all
right
very
well
so
I
like
to
I'm
tempted
to
repeat
that
well
I
think
that
it
provides
a
good
example
so
take
the
web
of
social
relation
or
social
events
that
we
may
want
to

model
so
this
provides
actually
a
good
nice
example
of
multi
scale
or
multi
scale
nature
of
target
phenomena
so
this
the
social
events
is
constituted
by
the
relationship
between
people
and
event
and
environment
and
well
people

are
actual
human
beings
they
are
biological
entities
constitute
of
people
who
are
self
organizing
system
and
we
are
able
to
explain
people
and
their
behavior
at
some
level
on
the
basis
of
biological
facts
so
we
may

be
able
to
use
some
sort
of
modeling
system
that
are
applicable
in
biology
or
that
sort
of
stuff
but
at
a
larger
scale
when
we
try
to
model
the
relation
between
different
people
as
nodes
in
networks

and
that
sort
of
stuff
we
may
need
to
change
our
modeling
device
to
I
don't
know
devices
that
are
good
at
representing
a
small
world
networks
or that
sort
of
stuff
if
that
does
make
any
sense
I

think
that
I
just
get
confused
by the
point
about
low
dimensionality
of models
in
social
systems
but
it
could
be
a
stevenism
or
one
way
at
least
that
I
would
read
that
would
be
like
the
maps
that
we

make
are
lower
dimensional
than
the
territories
they have
a different
resolution
but also
they don't
consider
they're not
trying to be
a full
inventory
of every
variable
in the
system
like
there
might
be
a
thousand
chemicals
circulating
in
our
blood
but
then
somebody
might
make
a

lower
dimensional
model
with
just
blood
sugar
or
two
dimensional
model
with
just
cortisol
and
blood
sugar
and
I
think
it's
an
active
inference
open
question
when we
make
lower
dimensional
models
of
complex
multi-scale
target
phenomena
like
you

addressed
it
I
mean
I
don't
want
to
jump
to
an
answer
but
it
seems
clear
like
that
the
philosophical
implications
of a
lower
dimensional
map
shouldn't
be
confused
with
a
systems
accounting
of
the
territory
because
it's
not

the
same
thing
and
yeah
Steven
just
followed
up in
the
chat
he
just
said
generally
how
does
Majid
see
the
informational
or
physical
sense
of
systems
like
some
people
talk
about
systems
in
the
more
ontological
sense
about

the
parts
and
how
they're
connected
other
times
we're
talking
about
systems
informationally
like
our
analysis
of the
measurement
data
and so
the
system
like
the
default
mode
network
is a
connectivity
pattern
that
isn't
necessarily
the
anatomical
connection
so
how do

we
think
about
these
different
kinds
of
systems
and
maybe
where
does
the
FEP
come
into
play
with
!
all
right
very well
that's
a
very
good
question
so
I'm a
tagbearer
structural
realist
so
I'm
committed
to
some
sort

of
information
theoretic
ontology
and
I think
that
well
and
I
say
that
I'm
defending
some
sort
of
embodied
informational
structural
realism
so
when I
say
that
I
was
advocating
cognitive
realism
it
could
have
been
more
precise
to
say

that
I'm
advocating
cognitive
structural
realism
which
is
actually
the
title
of
the
book
that
I
wrote
about
this
topic
so
and
what
rule
free
energy
principle
plays
in
this
context
is
a
very
complex
rule
so
let's

assume
that
we
have
the
structure
of
scientific
theories
or
scientific
models
on
the
one
hand
so
let's
assume
that
we
have
a
class
of
models
or
scientific
models
on
the
one
hand
and
on
the
other
hand

we
have
our
causal
structure
of
the
world
which
is
our
target
system
something
that
we
are
trying
to
represent
by
our
scientific
model
so
when
I
speak
about
a
structural
realism

I
think
that
it
has

been
the
official
wisdom
of
philosophy
of
science
for
the
past
15
years
or so
and
people
were
trying
to
use
different
kind
of
formal
vehicles
such as
set
theory
model
theory
category
theory
to
speak
of
this
representation
or

relation
between
the
class
of
models
and
the
causal
structure
of
the
world
right
but
my
beef
with
that
was
that
the
formal
relations
such
as
isomorphism
do not
contribute
to
strengthening
realism
because
these
formal
models
such
as

isomorphism

are

not

important

to

I mean

you

can

use

isomorphism

to

relate

any

two

sets

together

provided

that

there

have

enough

number

of

set

members

so

it

doesn't

contribute

to

realism

much

so

what

I

was

trying

to

do
was
to
use
free
energy
principle
to
account
for
the
class
of
models
and
their
target
system
in
the
world
assuming
that
scientists
are
self-organizing
system
that
aim
at
minimizing
the
discrepancy
between
their
models
and
their

target
system
in
the
real
world
so
but
it
is
my
take
I
mean
it
is
something
that
I'm
trying
to
develop
I
won't
I
don't
this
is
a
very
well
this
is
not
the
orthodox
view
either

in
the
philosophy
of
science
community
or
in
the
FIB
community
but
I
think
that
when
we
speak
of
representation
in
terms
of
the
representational
capacity
of
models
in
terms
of
the
relation
between
class
of
models
and

causal
structure
in
the
world
free
energy
principle
can
play
a
very
subtle
and
specific
rule
there
maybe
we're
in
a
little
strange
neck
of
the
woods
but
I
think
having
scientists
as
real
entities
doing
something
seems

at
home
here
blue
so
I
think
you
presented
us
with
a
lot
of
things
that
don't
contribute
to
realism
but
is there
anything
that
you
think
does
contribute
to
realism
well
could
you
elaborate
to
this
because
of

the
model
relativity
or
because
of
free
energy
principle
well
so
in
terms
of
cognitive
structural
realism
what
does
provide
evidence
for
the
cognitive
structural
realist
argument
I mean
you've given
us a lot
of things
that
you've
deconstructed
that
but
can you
take

the
opposite
side
and
provide
evidence
for
cognitive
structural
realism
or
do you think
it's
fair enough
look
the
general
I think
that the
general
arguments
for
scientific
realism
in general
have been
always
stated in
some
sort of
no miracle
arguments
assuming that
if
science was not
representing
the structure
of reality

it would
be a
matter
of
cosmic
co
the
empirical
adequacy
of
science
has been
a matter
of
cosmic
coincidence
so there
are
some
sorts
of
arguments
usual
arguments
in the
philosophy
of
science
that
try
to
relate
empirical
adequacy
of
scientific
theories
to

approximate
truth
of
scientific
theories
right
and
structural
realism
just tries
to
model
this
representation
of
relation
the
veracity
of
or
veridical
representational
relations
in terms
of
set
theory
and
model
theory
so
in
my
view
it
would
be
more

realistic
to
consider
scientists
as
actual
human
beings
constituted
by
flesh
and
blood
who
aim
at
minimizing
the
discrepancy
between
their
models
and
the
world
I
would
say
that
the
evidence
that
I
could
provide
for
this
take

on
realism
could
be
has
its
roots
in
evolutionary
biology
because
if
we
had not
been
successful
in
minimizing
the
discrepancy
between
our
model
and
reality
we
would
also
fail
to
maximize
our
survey
well
so
I
think
that

I
can
as
a
way
of
evidence
or
argument

I
can
try
to
groan
the
argument
the
usual
scientific
realist
argument

that
aim
to
relate
the
empirical
adequacy
of
science

to
truth
of
theory

I
can
try
to

relate
the
or
success
to
maximizing
our
survival
as an
evidence
of
or success
in
minimizing
the
discrepancy
between
our
scientific
models
and
their
target
system
the
causal
structure
of
reality
!
doesn't
require
a
specific
scale
of
analysis
which

is
what
you
brought
up
with
Markov
blankets
and
any
specific
application
will
entail
a
specific
scale
of
analysis
because
it's
empirical
and
so
it's
this
one
foot
in
both
worlds
just
like
linear
models
are
the
model

doesn't
have
a
priori
scale
but
then
when
we
specify
it
we
do
make
it
specific
and
what's
interesting
about
act
inf
is
that
we
could
think
about
this
action
and
perception
loop
of
different
kinds
of
nested

entities
now
definitely
it's
going to
get
complex
for
any
nested
entity
and
then
it
brings
a
whole
other
level
with
a
strange
loop
and
the
meta
modeling
and
the
cognizer
and
the
thinking
about
oneself
that's
like
a

whole
other
level
but
just
a
way
to
talk
about
vertically
and
horizontally
interacting
systems
is
an
awesome
start
so
blue
and
then
either
Majid
you
could
respond
to
what
blue
says
or
Dave
feel
free
to
say

hello
I
just
wanted
to
compliment
you
on
that
awesome
prelude
to
next
week's
discussion
about
modeling
ourselves
right
so
it's
a
very
interesting
perspective
when
the
modeler
is
the
model
it's
a
loop
cool
Dave
would you
like

to
say
hello
or
ask
anything
or
hi
nothing
particular
to
ask
sorry
to
be
late
I've
been
marinating
myself
in
neuropsychology
analysis
for the
last week
and a
half
that's
my
post
I've
got to
an
excuse
for
being
late
fair

enough
so
let's
go to
a few
of the
other
topics
that
we
had
written
down
so
I
think
before
we
get
to
this
more
the
the
liftoff
on
the
post
dot
two
let's
talk
a little
bit
about
intransitivity
because
in our

discussions
we had
looked at
intransitivity
in a
preference
framework
where
intransitivity
of preferences
has been
transformational
in
microeconomics
for example
and then
there's
intransitivity
from a
game
theoretic
perspective
like the
rock
paper
scissors
and just
strategic
intransitivity
which also
brings us
back to
the
population
biology
modeling
but
you

were
discussing
intransitivity
in a
specific
context
which is
about
the
predictive
processing
architectures
and
models
of
cognitive
architecture
so
could
you
maybe
unpack
a little
bit
what
does
intransitivity
mean
in that
context
who
disagrees
what
would
be
different
if
it

weren't
that
way
yeah
sure
so
I
think
that
the
thing
is
that
if
we
assume
that
there
is
transitivity
information
could
be
transferred
from
one
unit
to
another
we
would
have
a
serious
problem
in
defending
I

was
thinking
of
saying
something
more
about
active
inference
and
objective
inferences
because
I
will
put
that
on
the
background
for
the
time
being
but
the
thing
is
that
the
point
that
I
was
trying
to
make
in

the
paper
was
that
all
of
this
talk
about
the
distinction
between
transitivity
and
intransitivity
only
works
with
regard
to
invoking
a
specific
kind
of
models
which
is
no
wonder
because
the
paper
is
about
the
significance
of

scientific
models
so
we
can
defend
transitivity
or
intransitivity
of the
collective
structure
depending
on
the
kind
of
modeling
that
we
summon
so
if
we
assume
that
the
only
kind
of
models
that
are
allowable
in the
context
of
pre-energy

principle
are
dynamical
causal
models
we
will end up
by
a
straightforward
rejection
of
modularity
but
this
doesn't
need
to
be
the
case
because
pre-energy
principle
is
not
all
about
representing
the
actual
causal
mechanisms
neurophysiological
mechanisms
of the
brain
so

this
is
as
much
about
representing
this
neurophysiological
structures
as
the
general
properties
of
the
it
is
as
much
about
representing
the
local
significant
neurophysiological
mechanism
as it
is
about
explaining
the
relation
between
different
units
and
different
component

counterparts
that
are
implemented
in the
brain
so
it
is
supposed
to
be
a
unifying
theory
of
the
brain
and
in
some
sense
it
may
indicate
that
we
do
not
need
modularity
enough
because
if
we
want
to
defend

some
sort
of
unity
between
different
components
between
different
parts
of
the
structure
we
do
not
need
to
be
able
to
modulate
the
brain
and
the
paper
tries
to
argue
the
reverse
that
speaking
of
the
connection
between

different
parts
in
terms
of
mainly
in
terms
of
and
again
it
comes
back
to
the
distinction
between
effective
functionality
and
functional
connectivity
that
you
discussed
very
well
most
probably
better
than
I
would
be
able
to
in

the
previous
session
would
be
a
matter
of
the
kind
of
properties
of
the
collective
system
that
we
aim
to
model
by
invoking
free
energy
principle
whether
there
are
local
properties
of
the
brain
or
the
global
unifying

properties
of
the
brain
and
if
we
want
to
model
the
brain
in
its
unity
we
also
need
to
be
able
to
speak
about
the
connection
between
different
parts
and
from
the
very
fact
that
we
assume
that

there
are
different
parts
it
follows
that
we
have
to
assume
that
there
is
some
sort
of
encapsulation
or
a
weak
point
a
moderate
form
of
modularity
very
interesting
thank you
any other
thoughts
on that
well then
I'm going
to go
to a
hard

question
which
is
what
makes
a
model
active
inference
is
that
a
in
the
in
the
circle
or
out
of
the
circle
is
it
a
binary
classifier
is
it
a
zero
to
one
scale
what
will
happen
when

people
disagree
does
simply
any
model
involving
action
and
inference
qualify
as
active
inference
we
have
asked
it
on a
few
previous
discussions
and I
think we
have
gotten
a range
of
answers
but it's
definitely
an
important
one
to
revisit
from
this

model
based
science
perspective
well
I mean
I have
been
defending
some
sort
of
panpsychism
I have
been
marshalling
free energy
principle
in defense
of
versions
of
panpsychism
so
I'm tempted
to say
that
any
kind
of
self
organizing
system
that
can
minimize
the
discrepancy

between
its
internal
models
and
the
environment
is
capable
of
forming
active
inferences
but
it is
not
directly
to
I mean
it is
not
specifically
related
to
this
issue
of
model
relativity
unless
we assume
that
model
relativity
is
quite
ubiquitous
it is

everywhere

in

scientific

activity

I

cannot

find

any

specific

relation

between

that

kind

of

panpsychist

account

of

active

inference

and

a

specific

discussion

of

model

relativity

Daniel can I

ask a

question

yep

Dean

and

sorry

Blue

maybe you

want to

go first

you put

your
hand
up
and
I
just
blurt
it
out
go
ahead
Dean
okay
sorry
about
that
thanks
so
at
the
end
of
the
last
live
stream
there
was
a
question
posted
in
the
live
chat
about
so
does

active
maybe
this
ties in
with
what
you
were
just
asking
Daniel
does
active
inference
tell
us
about
the
model
of
the
person
trying
to
kick
the
ball
and
I'm
I'm
curious
about
this
because
you
wrote
another
paper

with
a
you
co-authored
a
paper
on
active
inference
and
abduction

with
somebody
if I
pronounce
their
last name
incorrectly
I apologize
Pia
Terranin
and
I'm
kind of
curious
because
I think
what
active
inference
and
active
inference
within
model
building
and

modularity

is

it

it

it

it

gets

us

asking

a

question

much

like

Charles

Saunders

Pierce

would

ask

if

you

walked

in

late

on

Bob

and

Alice

okay

it's

nine

o'clock

but you're

not really

paying

attention

to the

clock

right

now
you're
just
walked
into
a
room
and
there's
on
the
other
side
of
the
one
way
mirror
is
a
subdivided
two
cubes
one
with
Bob
in
and
one
with
Alice

the
curtain
you
don't
have
a

smiley
face
or
a
frowny
face
what
you
have
is
a
moment
where
you
have
to
ask
if
you're
actively
inferring
and
you're
enacting
this
is
what
should
I
be
surprised
if
and
what
should
I
not
be

surprised

if

as

the

as

the

as

the

stuff

that's

carrying

out

in

Bob's

room

somehow

is

appears

to

not

being

carried

out

in

Alice's

room

so

I

don't

know

that

modeling

can

sort

of

segregate

itself

out

scale
free
from
the
idea
of
the
specifics
like
Pierce
would
say
if
you
walked
in
into
a
room
and
somebody
opposite
of
you
had
a
bunch
of
white
beans
in
their
left
hand
and
a
bag
of

beans
in
their
right
hand
and
asked
you
as
the
person
walking
to
the
room
did
these
beans
in
this
one
hand
come
from
this
bag
you
wouldn't
have
enough
information
to
be able
to
give
a
specific
and

clear
answer
to
that
but
you
could
start
a
process
of
modeling
you
could
start
a
modularity
exercise
to
try
and
gather
from
an
abductive
sense
what
led
up to
this
moment
and
so
I'm
just
kind
of
wondering

Majid
how
do
you
take
this
paper
and
that
example
and
sort
of
tie
it
in
with
some
of
the
other
work
that
you've
been
because
I'm
hoping
maybe
I'll
talk
to
you
after
this
but
maybe
I

would
like
to
be
able
to
review
your
active
inference
and
abduction
paper
as
well
if
you
have
time
anyway
I'm
just
wondering
what
because
I
think
if
we're
looking
at
this
from
active
inference
and
enacting
sometimes

we
walk
in
and
we
walk
in
and
we
have
to
infer
based
on
a
limited
amount
of
information
and
so
I
think
that's
what
we're
trying
to
do
we're
basically
saying
when
we
walk
in
do
or

don't
be
surprised
if
and
then
whatever
plays
out
approximately
on
those
time
scales
next
yes
I
do
agree
and
well
I'm
very happy
that you
mentioned
that
paper
because
I
wanted
to
speak
about
that
too
earlier
so
the

thing
is
that
as
I
say
there
are
two
phases
of
the
same
project
so
in
this
project
in
the
project
that
has
been
stated
in
this
tale
of
two
architectures
and
in
another
paper
a
critical
artist

of
Markovian
monism
I
was
emphasizing
the
model
base
model
relative
nature
of
active
inference
as
an
activity
but
the
main
reason
why
I
was
getting
interested
in
that
because
of
the
Markov
blankets
because
I
think
that

Markov
blankets
are
also
very
strong
and
reliable
modeling
tools
also
for
the
general
scientific
modeling
purposes
so
it
is
the
other
side
of
my
project
when
I
trying
to
groan
scientific
realism
and
scientific
model
making
in

active
inference
and
free
energy
principles
so
this
is
the
other
side
and
going
back
to
the
general
arguments
for
scientific
realism
again
a strong
version
of
defense
of
scientific
realism
is
reliant
on
invoking
explanatory
inferences
by
developing

something
that
is
called
inference
to
the
best
explanation
and
this
project
that
you
mentioned
the
paper
that
has
been
written
by
me
and
Ahti
and
there
is
another
paper
we
were
collaborating
on
another
paper
on
the

same
subject
aims
to
show
that
how
the
general
explanatory
inferences
how
the
general
abduction
is
grounded
in
the
process
of
active
inferences

right
so
I
think
that
there
are
two
sides
of
the
same
coin
and

they
are
two
intertwined
two
interrelated
projects
but
as I
say
I
do
not
see
so
do
you
want
to
also
explore
the
implications
of
that
for
the
metaphor
of
Bob
and
Alice
I
mean
do
you
think
that

it
has
any
specific
ramifications
for
this
Bob
and
Alice
metaphor
yeah
that's
that
was
that's
exactly
what
I
think
it
does
and
again
it's
a
separate
paper
and
another
excellent
excellent
insight
if you
get a
chance
to read
it

but
I
think
when
we're
trying
to
figure
out
what's
effective
and
functional
we
come
in
to
a
situation
looking
at
Bob
and
Alice
and
over
time
our
model
grows
as
well
right
and
I
think
that's
part

of
where
my
question
about
what
is
what
is
scale
free
yes
Bob
and
Alice
are
in
their
cubicles
and
those
cubes
are
of
the
same
dimensions
and
then
there's
a
scale
friendly
part
that
kind
of
grows

over
those
slices
of
time
so
yes
I
actually
do
think
that
it
would
be
interesting
to
sort
of
say
when
did
we
walk
in
on
this
Bob
and
Alice
observation
very
well
so
I
would
say
that

for
the
moderate
weak sense
of
encapsulation
that is
at issue
in this
specific
paper
the tale
of
two
city
it
would be
the
necessary
condition
is that
the
information
in each
unit
is not
available
to
the
person
in
the
other
unit
or
in
the
other

black
so
to
the
extent
that
the
information
about
for
example
Bob
brushing
his
is
not
available
to
Alice
there
is
some
sort
of
weak
encapsulation
and
I
think
that
it
is
enough
for
establishing
some
sort
of

moderate
modulation
but
the
other
part
the
more
fundamental
question
about
the
scale
relativity
and
to
what
extent
so
it
is
some
sort
of
scientific
fact
that
we
can
choose
the
scale
that
we
need
on
the
basis

of
our
explanatory
interest
so
at
this
specific
setting
my
explanatory
interest
my
scientific
interest
was
to
set
the
limits
of
the
scale
at
the
level
of
the
rooms
of
Bob
and
Alice
because
I
wanted
to
explain

encapsulation

so

in

another

situation

so

assume

that

you

are

an

information

theorist

and

you

want

to

develop

a

general

theory

of

communication

between

these

two

units

so

it

is

natural

for

you

to

take

the

point

of

view
of
the
third
observer
and
then
there
might
be
another
person
who
wants
to
speak
about
the
relation
between
this
third
person
and
the
target
system
and
it
might
go
observer
all the
way
up
and
observer
all

the
way
down
and
it
would
be
a
matter
of
what
you
are
trying
to
do
with
changing
the
scale
what
goal
do
you
aim
to
achieve
so
again
at
some
papers
I
was
trying
to
stay
limited

to
the
level
of
individual
so
for
example
when
I
wanted
to
model
the
activity
of
a
scientist
specific
and
individual
scientist
I
was
focusing
on
the
processes
of
minimization
of
predictive
predictions
at the
level
of
individuals
but

at
points
I
wanted
to
model
the
behavior
or
speak
about
the
behavior
of
all
scientific
commodity
assuming
that
scientific
practice
is
a
social
act
and
I
was
for
that
reason
I
was
trying
to
expand
the
limits

of
the
scale
and
apply
that
to
the
old
commodity
of
scientists
or
a
specific
group
of
scientists
rather
than
individual
scientists
what
was
important
for
me
and
what
I
think
is
important
is
to
have
good
and

satisfactory
justifications
for
changing
the
level
to
be
able
to
explain
clearly
what
was
the
purpose
of
changing

the
level
!
taking
it
to
an
upper
level
or
a
lower
level
and
I
think
that
to
the

extent
that
we are
speaking
about
encapsulation
and modulation
it is
justifiable to
limit the
module to
the specific
cells in
which Bob
and Alice
are dwelling.
So just
wrote a few
things down
and maybe
this captures
it but
the free
energy
principle
especially
as you're
applying it
in this
embodied
informational
and cognitive
structural
realism
perspective
the FEP
as a
theory

of
things
is
providing
the
conceptual
continuity
for
modeling
different
complex
systems
including
scientists
in the
world
and other
processes
as they
engage
in their
niche
and
then
the
active
inference
framework
we can
say
is
operationalizing
this
imperative
for
persistence
of
different

systems
and
then
connecting
that
a
priori
process
ontology
to
a whole
host
of
other
patterns
motifs
that we
must
explain
in the
world
like
abduction
in the
social
and
communication
case
and
narrative
information
inference
on one
end
of us
but then
on the
other

end
we also
want
to have
act
dimp
really
dial
in
and
get
at
exact
inference
in
certain
cases
where
it's like
we made
the
fMRI
measurement
and we
kind of
want to
leave
philosophy
in the
lab
and just
plug and
chug
this
pipeline
and we
don't want
that to be

a pipeline
of nowhere
or we
don't want
the output
of the
pipeline
to be
interpreted
in a
fallacious
philosophical
milieu
but at
the same
time
if we
try to
take that
philosophical
baggage
through
every
step
of the
processor
and the
RAM
and the
GPU
we get
into these
little
nano
debates
which you
highlighted
which is

like
people
arguing
whether
cyclic
or
directed
acyclic
graphical
models
have
philosophical
impact
when
they're
both
just
models
that
scientists
chose
for
different
reasons
I have
to say
that you
have the
knock of
explaining
my ideas
far more
clearly
than I
can
so
here's
my next

question
for you
because
there's
multiple
FEP
frontiers
here
you have
the
book
then
there's
the
Markovian
monism
that we
had earlier
discussed
the
abduction
one
which we
haven't
gotten a
chance
to
discuss
this
paper
obviously
so
how do
you keep
the
information
organized
or

assess
how
new
papers
coming
in
play
into
how
you
want
to
develop
these
different
threads
of
research
or
determine
where
you
can
be
making
an
impact
by
targeting
this
paper
or
assessing
this
phenomena
or
stepping
into

this
5,000
year
old
philosophical
debate
well
it is
very hard
to
explain
where
these
ideas
are
coming
from

!
simpler
bits
so
for
example
starting
by
just
stating
the
point
that
scientific
representation
could be
grounded
in
free

energy
principle
and
then
I'm
just
fleshing
out
the
other
complex
issues
so
for
example
one
thing
so
as
they
say
one
thing
leads
to
another
as
soon
as
you
try
to
account
for
scientific
representation
in terms
of

free
energy
principle
people
pointed
and there
have been
excellent
papers
by
excellent
criticism
for
example
by
Max
Jones
who
were
pointing
out
this
criticism
that
science
is not
an
individual
activity
it
is
a
social
activity
and
then
I
am

going
back
to
the
social
aspects
of
my
view
so
I
think
that
it
is
again
some
sort
of
dynamical
social
process
that
leads
to
development
of
these
ideas
and
then
I
am
trying
to
deal
with
this

social
aspect
I
stumble
on
reading
Thomas
Kuhn
and
I
see
that
I
have
to
also
take
care
of
theory
relatedness
for
example
and
I
engage
with
that
issue
but
I
think
that
I
started
to
work
on

this
applying
this
model
relativity
the
model
based
nature
of
scientific
activity
to
philosophy
of
mind
since
2016
and
then
as
I
say
one
thing
led
to
another
and
I
am
where
I
am
now
and
I
have

no
idea
of
what
impact
would
this
project
lead
to
what
group
of
people
because
most
probably
the
project
is
a bit
too
scientific
for
philosophers
and
a bit
too
philosophical
for
scientists
but
well
it
is
the
only
way

that
I
know

for
developing
my
philosophical
idea
and

I
think
you
blue
if
you
wanted

to
raise
your
hand
otherwise

go for
it
if

not
then
okay

blue
then

Dave
so

I
think
my
comment
was
it's

kind
of
old
at
this
point
but
but
really
like
I
was
just
curious
about
linking
this
or
or under
what
circumstances
or how
can
we
apply
this
the
active
inference
framework
to
partially
encapsulated
systems
and so
I mean
I think
that this

is
it gets
into
like
I guess
that's
the big
question
right
so
always
just
I would
love
to hear
your
thoughts
on
how
information
is
passed
from
module
to
module
or
what
is
this
idea
of
mutual
information
you
know
in
the

collective
panpsychism
sense
like
how
do
you
think
or
to
what
degree
is
it
shared
and
can
you
perhaps
maybe
shed
some
light
on
mathematically
ways
that
you
think
it
would
make
sense
to
talk
about
partial
encapsulation

well
so
starting
from
the
last
part
I
think
that
mathematical
representation
is
rather
the
easier
part
so
we
can
easily
model
this
partial
encapsulation
by
invoking
some
sort
of
partial
isomorphism
so
it is
enough
to
have
two

sets
of
models
and
try
to
show
that
to
what
extent
or
what
relation
what
parts
what
sub
parts
of
these
sets
are
related
to
what
subsets
of
the
other
system
to
the
extent
that
there
are
significant

chunks
that
are
not
related
together
through
some
sort
of
isomorphic
relation
we
can
retain
some
sort
of
partial
isomorphism
the
other
way
I
think
that
the
more
intuitive
way
of
doing
that
is
by
invoking
Markov
blankets

I
think
that
as a
matter
of
fact
I
think
that
partial
encapsulation
is a
direct
result
ramification
of
invoking
Markovian
blankets
because
they
are
setting
some
sort
of
conditional
independence
between
different
chunks
right
so
well
from the
very
fact

that
there
is
some
sort
of
partial
encapsulation
it
doesn't
follow
that
there
is
no
amount
of
transfer
of
information
between
the
chunks
at all
the
only
thing
that
is
necessary
for
retaining
this
partial
encapsulation
is that
at
a

given
moment
there
is
no
actual
track
so
the
only
thing
I
think
that
the
necessary
condition
and
it
is
very
minimal
for
retaining
some
sort
of
partial
encapsulation
is that
at
any
given
time
information
at
one
unit

is
not
accessible
to
the
observer
in
the
other
unit
and
I
think
that
this
condition
could
be
satisfied
quite
easily
it
is
why
I
think
that
some
sort
of
partial
encapsulation
endures
despite
the
validity
of
Kirchhoff

and
Hippolyto
arguments
thanks
blue
and then
go ahead
so
so I
guess
this
is
how
so
my
question
was
probably
backwards
so what
about
how
is
information
shared
not
is
how
do
you
demonstrate
information
sharing
as opposed
to
information
encapsulation
like

if
there's
a
Markov
blanket
we
establish
conditional
independence
but
like
where
can
we
establish
conditional
dependence
or
how
can
we
establish
conditional
dependence
that's
the part
that's
really
I'm
curious
about
sorry
all right
very well
so
look
the
internal

and external
states
or the
sensory
and active
states
are related
together
but the
thing
is
that
the
observer
or
if
you
are
encapsulated
inside
the
Markov
blanket
you
can
infer
the
properties
of
outside
only
indirectly
so
what I
was
trying
to do
to some

extent
is to
instead of
assuming
that the
collective
system
is composed
of
Markov
blankets
I
tried to
put
myself
inside
the
Markov
blanket
and
see
how
could I
get
access
to
information
that
is
outside
the
Markov
blanket
or
inside
another
Markov
blanket

and
to the
extent
that
there
is
no
direct
relation
or
the
information
is not
directly
accessible
at a
given
time
I
think
that
some
sort
of
encapsulation
induce
like
a
cup
half
full
cup
half
empty
channel
half
open
channel

half
closed
it's
the
two
extremes
!
Complete
integration
which
maybe
doesn't
even
make
any
sense
like
a
solid
state
drive
I
mean
what
would
it
even
mean
if
all
the
information
were
connected
to
all
the
other

information
at
the
same
time
and
just
not
compatible
with
the
architectures
of
physical
systems
so
interesting
thought
experiments
though
for
total
integration
and
then
any
realized
system
is
either
going
to
be
fully
conditionally
independent
entirely
Markov

blanketed
or
there'll
be some
partial
relation
and
then
part
of
the
realism
and
instrumentalism
okay
that
partial
relation
can
be
an
edge
like
an
effective
or
functional
connectivity
edge
which
doesn't
imply
a
causal
role
it's
a
scientific

projection
situationally
defined
edge
that's
being
inferred
in a
specific
statistics
framework
and then
there are
hidden
states
of the
world
of
observers
who
are
supposing
causes
like
hypotheses
about how
things happen
in the
world
and then
they engage
in all
kinds of
studies
like
loss
of
function

gain
of
function
blinded
research
those
are
protocols
or
affordances
that
scientists
engage
in
to reduce
their
uncertainty
about
this
hypothesis
space
so it's a
very
interesting
area
Dave
and then
Dean
right
it's
perfect
lead
into
what I
was
going
to
ask

about
the
where
you
draw
the
boundaries
where
you
draw
the
blankets
is
really
important
I like
to say
nasty
things
about
Donald
Hoffman
or more
specifically
the way
he allows
himself
to be
portrayed
as
thinking
that
there's
absolute
freedom
in
throwing
blankets

wherever
you
want
that
the
boundaries
are
totally
arbitrary
and
so
we
are
all
one
really
though
blanketing
is a
matter
of
where
what's
actually
happening
are you
Majid
familiar
there we
go
are you
familiar
with
Mara
Beller's
work
on
the

level
of
actual
scientific
interaction
can
you
describe
that
work
a
little
bit
Mara
Beller
yeah
she
was
for
a
number
of
years
she
was
perhaps
the
leading
historian
of
the
early
days
of
quantum
mechanics
and
as she

got
more
and more
in
depth
into
what
actually
was
happening
there
are
certain
papers
by
Bohr
that
are
widely
regarded
as being
totally
incomprehensible

You go
from
one
thing
that
he's
saying
in
the
paper
to
a
few
sentences

later
and
there
doesn't
seem
to
be
any
connection
at
all
and
people
were
thinking
well
what
was he
had he
lost it
was he
too tired
when he
gave this
talk
well she
went down
and trying to
take sentence
by sentence
and phrase
by phrase
asking the same
question that a good
drama coach
asks
for each
statement

that you
have to
deliver
what
question
is that
speech
an answer
to
and she
found
if you
went through
the people
that
Bohr
was in
intimate
dialogue
with
whether he
was on
their side
or
being driven
crazy
by them
like
Einstein
if you
kept that
in mind
and asked
who was he
talking to
who was
Bohr
talking to

at each
moment
everything
became
very
straightforward
he was
having
a series
of
dialogues
with
a dozen
people
and just
stringing
one fragment
of the
dialogue
after another
and I
haven't read
it yet
but that
totally ties
in with
the concept
of
conversation
theory
which was
something
put together
in the
1960s
and 70s
by my
teacher

of
cybernetic
learning
theory
Gordon
Pasque
who
defines
the
minimum
unit
of
consciousness
as
the
conversation
and also
represents
each
conversation
or at least
each
enduring
conversation
as a
psychological
individual
so you
have very
much
nested
mark
blankets
!
You have
very much
the
Leibnizian

hierarchy
and the
indefinitely
large or
infinitely
large and
deep
hierarchy
of
minds
composed of
minds
composed of
minds

and
the way
Beller
presents
that
that's
what
science
is
about
having
lots
of
dialogues
figuring
out
how
they
fit
together
the
dialogues
begetting

new
dialogues
and
in
past
sense
those
conversations
themselves
in the
best
case
become
the
enduring
individuals
maybe
the
immortal
individuals
that
make
up
a
successful
living
science
Thank you
very much
for
the
input
I agree
that
the
issue
of
choosing

the
scale
is not
arbitrary
so I
think
that
earlier
in the
paper
I
referred
to
the
works
of
Rodolf
Carnap
and
Thomas
Kuhn
to
point
out
that
I
think
that
there
should
be
some
general
considerations
that
rule
over
the

choice
of
a
specific
scale
this
is
partly
determined
by
the
property
of
the
target
system
itself
and
partly
by
the
interest
of
the
modeler
depending
on
what
the
modeler
is
trying
to
work
out
about
the
target

system

so

I

completely

agree

with

that

and

about

!

the

collaborative

and

distributed

so

I don't

know

whether

it is

I'm

reading

too

much

into

what

you

said

right

but

I'm

completely

on board

with

some

sort

of

collective

distributed

take
on
scientific
activity
and
this
is
something
that
I
have
been
defending
in
recent
papers
in
the
previous
year
so
I
completely
agree
with
that
and
well
my
so
the
point
about
individuals
and
how
to
define

an
individual
or
the
borders
of
consciousness
the
limits
of
consciousness
is
still
very
much
at
the
center
of
my
intellectual
enterprises
so
this
is
something
that
I'm
trying
to
explore
more
so
and
this
is
something
that

I'm
thinking
about
these
days
whether
the
unit
of
scientific
activity
is
a
region
in
the
population
of
neurons
in
the
brain
of
scientists
or
an
individual
scientist
or
a
group
of
scientists
so
these
are
all
very

interesting
topics
thank
you
very
much
thanks
for
the
response
so
dean
and
then
blue
so
I think
what
you've
done
is
you've
hit
on a
really
important
point
and that
is that
whenever
we do
isomorphism
!
we can
sort of
take
an
iso

an
iso
stance
I've
read a
book
a long
time
ago
by
the
editor
author
was
James
Brockwell
and it
was
called
The
Visual
World
and Memory
and what
he
said
was
is
that
the
way
that
we
capture
can
be
one
of

two
types
the
isovisstance
that
sort
of
isomorphism
that
we're
outside
and we're
creating
a
representation
with
frames
around
them
or
we
can
be
spatially
enveloped
which
kind
of
ties
in
really
conveniently
with
the
idea
of
being
either

in
the
cubbyhole
that
Bob
is
in
or
Alice
is
in
or
anybody
that's
within
that
sort
of
entailed
world
where
we're
sort
of
now
the
subject
of
whatever
the
walls
are
limiting
us
from
being
able
to

see
and
so
I
don't
think
we
walk
away
from
the
idea
that
we
can
make
models
we
just
have
to
be
able
to
ask
on
a
moment
especially
to
to
Blue's
point
how
do
we
make
something

available
as
opposed
to
keeping
it
sort
of
cleaved
off
or
sectioned
off
we
have
to
kind
of
take
the
perspective
or
the
view
of
the
person
who
is
enveloped
who
doesn't
necessarily
have
that
eyes
of
this

slash
model
ability
to
nest
thing
and
look
at
the
world
from
their
eyes
which
is
okay
it's
not
what
you're
seeing
necessarily
in terms
of
what's
on
the
page
it's
how
you're
entering
the
room
with
maybe
not

as
much
background
and
not
as
much
as
ability
to
step
outside
of
all
of
this
information
and
start
juggling
and so
I think
that
is one
of the
really
key
points
to
try
to
remember
here
when
we
model
that
just

because

I

have

a

model

that

I

can

take

that

eyes

of

a

stance

doesn't

mean

I

am

able

to

also

remember

what

it

was

like

to

be

the

one

sort

of

captured

or

catch

up

I

think

if
we
can
hold
both
up
at
once
it's
back
to
that
minimum
two
things
at
once
don't
be
surprised
if
do
be
surprised
if
or
active
and
inference
like
I
think
what
we're
really
basically
saying
here

is
bumping
it
up
by
including
two
things
that
sort
of
depth
perception
piece
yes
it
does
make
it
more
complicated
or
complex
but
that
doesn't
mean
it's
overwhelming
it
dimensions
at
once
but
we
can
bump
up

to
two
we
can
take
the
eyes
of
a
stance
we
can
do
the
model
and
we
can
also
remember
what
it
was
like
to
be
spatially
enveloped
you're
going to
get
overwhelmed
and
surprised
by
the
real
world

when
the
train
comes
at
you
if
you
don't
have
depth
perception
so
it's
a
very
interesting
take
Dean
like
yes
two
and
two
is
truer
than
one
but
also
there's
other
benefits
too
blue
so
just
to

kind
of
clarify
like
what
I
was
talking
about
earlier
and
then
add
on
to
what
Dave
was
saying
like
I
want
to
know
how
to
draw
a
partial
Markov
blanket
like
how
do
you
like
to me
a

Markov
blanket
represents
a
distinct
boundary
between
self
and
other
internal
and
external
and
so
how
do
you
make
this
partial
Markov
blanket
and
I
like
what
Dave
was
saying
or
about
what
his
teacher
put
forward
about

the
conversation
being
a unit
of
consciousness
and
I
am
very
much
a
ascriber
to
panpsychism
like
I
don't
think
that
it
necessarily
needs
to be
a
conversation
between
two
humans
but
really
maybe
fundamentally
each
relationship
between
two
objects

perhaps
each
relationship
has
then
its
own
Markov
blanket
like
you
know
my
heart
and
my
liver
have
a
relationship
with
each
other
that
that
itself
is
encapsulated
with
a
Markov
blanket
and
this
can
be
attributed
to

any
two
cells
or
two
organisms
or
so
on
and
so
forth
very
well
so
by
the
way
of
an
example
I
completely
agree
so
especially
about
the
part
that
this
so
well
I
do
not
want
to

use
the
term
conversation
because
I
do
not
know
the
theoretical
implications
very
well
but
I
think
that
so
I
used
this
when
I
started
this
project
for
example
on
modeling
scientific
activity
as
some
sort
of
active

inference
I
was
thinking
of
individual
scientists
as
the
units
of
scientific
activity
so
an
individual
scientist
has been
enveloped
by
Markov
blanket
and
she
was
trying
to
get
access
to
the
causal
structure
of
the
world
by
forming

inferences

from

beyond

the

blanket

but

as

I

work

more

on

this

aspect

I

understood

that

the

situation

is

becoming

complex

very

quickly

so

then

I

told

that

we

could

put

all

of

the

scientific

group

or

scientific

community
inside
the
Markov
blanket
and
then
try
to
explain
their
relation
with
the
environment
as
the
parts
of
a
general
Markov
blanket
by
just
expanding
the
scale
that
includes
them
and
then
at
the
stage
I
also

told
that
we
could
also
model
in
the
same
way
that
we
could
envelope
the
relation
between
two
scientists
inside
the
Markov
blanket
we
could
just
expand
that
and
also
include
the
artifact
the
scientific
experimental
tools
inside

the
community
of
the
scientists
so
scientific
activity
is
basically
about
the
relation
between
scientist
scientist
human
human
activity
and
human
artifact
relationship
and
scientific
activities
distributed
to
all
of
this
group
which
includes
both
human
beings
and

artifacts
and
then
depending
on
your
take
on
free
energy
principle
whether
you
are
defending
a
representationalist
stance
or an
activist
stance
you
can
speak
about
the
relation
of
this
group
to
their
environment
in
terms
of
classical
representation

or
coupling
or
a
hybrid
view
which
includes
both
representations
and
coupling
thank you
very
interesting
makes me
think about
science
and
technology
and
how
the
word
science
relates
to
like
a lot
of
inference
and
sigh
and
scry
and
all
that

and
then
technology
technique
and
that
has a
little
bit
more
of
the
implications
of
tool
deployment
and
finesse
and
measurement
and
the
blurry
line
between
science
and
technology
and
the
who
and
the
what
and
the
how
so

it's
very
cool
let's
in our
last
little bit
here
first if
anybody
watching
live
has
questions
we
totally
can
address
them
but
this
was
the
question
that
we
left
off
on
at
the
end
of
the
point
one
video
which

was
how
do
you
utilize
functional
and
effective
connectivity
in
active
inference
analysis
to
better
inform
an
answer
to
a
question
such
as
in
football
why
I
scored
the
goal
so
how
does
that
enrich
in
analysis
because

we've
talked
a lot
about
how
the
chosen
models
deployed
by
scientists
are
goal
driven
and
so
what
exactly
is
active
inference
helping
us
do
or
how
is
it
enriching
our
account
of
a
situation
like
scoring
a
goal

in
a
sports
ball
game
well
as a
philosopher
I feel
that I
have to
stay
silent
about
this
question
to
avoid
revealing
my
deep
ignorance
of the
technical
aspects
of how
this
works
so
I
mean
I
have
some
general
ideas
and
I

can
try
but
I
think
that
it
would
be
better
to
stay
silent
and
not
reveal
my
ignorance
how
about
start it
off
with
something
ignorant
and
then
we'll
see
if
any
other
researchers
have
thoughts
well
I
do

not
know
about
this
specific
example
but
for
me
the
distinction
between
functional
connectivity
and
effective
connectivity
was
about
the
causal
physical
aspects
of the
target
system
and
the
general
global
and
structural
features
of the
system
so
I
wasn't

thinking
of
this
in
terms
of
so
the
point
that
I
was
trying
to
make
whether
to
try
to
focus
on
modeling
the
causal
physical
aspects
or
focus
on
the
structural
global
aspects
would
be
a
matter
of

the
interest
of
the
scientist
or
the
modeler
whether
at
the
moment
or
whether
in
order
to
do
the
task
at
hand
she
prefers
to
stay
focused
on
the
general
aspect
of
speaking
about
the
relation
between
different

parts
in
unifying
structure
terms
or
going
into
the
details
and
trying
to
model
local
specific
features
I'm
not
quite
sure
how
I
can
apply
that
to
the
example
of
scoring
a
goal
Dean
or
anyone
else
I

just
think
it's
in
scoring
the
goal
again
pre
pre
kicking
the
ball
you
could
be
asking
yourself
two
questions
why
am
I
surprised
about
the
result
or
should
I
be
surprised
about
the
result
because
I
think

that's
what
active
inference
affords
it
gives
you
that
possibility
of
two
outcomes
I
scored
a
goal
or
I
kicked
it
up
into
the
stands
but
before
you
actually
kick
the
ball
perhaps
perhaps
if
you're
not
really

skilled
at
kicking
a
ball
you
can
ask
yourself
whether
or
not
there's
enough
enough
in this
particular
case
mechanical
familiarity
and
background
to
actually
finish
off
what
you
intended
to
do
that's
what
I
think
active
inference
allows

for
but
again
my
ignorance
shines
through
too
so
okay
I'll
give
a
thought
so
I
think
there's
active
inference
and then
there's
functional
and effective
connectivity
because
functional
and effective
connectivity
have gone
back
further
so
first
applying
the
techniques
of

functional
effective
connectivity
help us
separate
out
hypotheses
about
physical
causal
mechanisms
from
a whole
variety
of
different
kinds
of
edges
that
might
be
applied
like
there's
time
correlation
there's
mutual
information
there's
change
with respect
to
like
you know
condition
on this

one
going up
in that
split
second
this
one
goes
up
as
well
so
all
kinds
of
different
edges
that
can
be
drawn
essentially
in a
graphical
framework
so
not
just
like
visual
but
on
nodes
and
edges
active
inference
gives

us
a
I mean
among other
things
so this
of course
would not
be a
complete
or
authoritative
answer
but it
does provide
us with
a framework
and a
nexus
and a
community
that can
scaffold
and nurture
accounts of
the target
phenomena
so
kicking
a
goal
we
want
to
have
the
biomechanics
researcher

and the
force
on the
knee
and
physical
rehab
and then
also
there's
an easy
way to
go into
like the
social
intersubjectivity
of the
game rules
and
regimes
of attention
and thinking
through other
teammates
and
cyber
physical
systems
of the
soccer
stadium
and then
like
reviewer
two
just
addressing
the

inner
or
outer
critic
and
knowing
that
we
don't
need
to
have
a
theory
of
everything
to
talk
about
the
ball
being
kicked
so
I'd
say
that
ACTAMP
specifically
provides
value
for the
researcher
the
goal
scoring
researcher
because

it will
help
them
reduce
their
uncertainty
about
what
contribution
to
make
to
the
research
literature
by
just
giving
really
good
connection
points
if
they're
doing
the
biomechanics
paper
it can
have
awesome
connection
points
to the
cultural
and the
social
and all

of that
and if
they're
doing
the
social
paper
it
could
have
nice
articulation
with
the
biomechanics
and that
would even
be before
we get
into
the
some
of the
details
of the
ACTIMF
modeling
per se
so

if
anyone
in
the
live
chat
would
like

any
final
questions
otherwise
we will
sort
of
cruise
out
you know
in the
dot
zero
we do
the
background
and
it's
the
time
that
we
all
need
to
take
to
work
through
these
very
provocative
and
interesting
and
far
reaching
papers

dot
one
is a
lot
of
times
like
a
mess
because
we're
exploring
a lot
of
different
avenues
and
we
never
finish
them
all
in
the
dot
two
but
this
is
sort
of
where
we
take
off
into
our
next

set
of
discussions
it
happens
to be
that
in
the
coming
weeks
we're
going
to
be
talking
about
the
nature
of
self
representation
so
that
might
have
some
themes
but
Majeed
I'm
just
curious
what
are
you
excited
about

for
2022
what
directions
or
policies
are
exciting
to you
now
that
you've
very
recently
completed
this
work
well
I'm
excited
about
a new
book
that is
under
review
and
it's
true
so
structural
realism
as I
said
was the
official
reason
of

philosophy

of

science

for

quite

a while

and

I

tried

to

develop

a

cognitive

version

of

structural

realism

before

in a

new

book

that

is

going

to be

published

it is

a sequel

to the

book

that you

are

seeing

in a

new

book

that

is

under
review
right
now
I'm
going
to
flesh
out
the
social
aspects
of
scientific
activities
so
structural
realists
despite
all
of
their
virtues
they
were
not
too
much
attending
speaking
about
the
social
aspects
of
science
and
I

think
has
been
neglected
in
the
contemporary
philosophy
of
science
so
it
had
its
high
day
in
works
of
Thomas
Kuhn
and
Feyerabend
and
then
it
has
been
pushed
into
the
background
and
now
I'm
building
upon
free

energy
principle
so
the
theories
of
social
communication
that
are
grounded
in
free
energy
principle
to
bring
back
the
discourse
on
social
aspects
of
scientific
activity
so
the
book
that
is
under
review
the
title
is
aptly
called

bringing
social
awareness
to
structural
realism
so
I'm
very
excited
about
that
I
still
do
not
know
whether
it
will
pass
the
review
and
whether
it
will
go
to
the
press
or
not
but
I'm
very
much
excited

about
that
nice
reminds
me
of
Bruno
Latour
reassembling
the
social
of
course
different
aim
but
that
almost
feels
invoked
in the
title
well
it is
a
different
aim
but
as I
say
it
has
so
it
is
mainly
about
works

of
Thomas
Kuhn
and
Pyroben
but
it
also
has
references
to
the
works
of
social
constructivists
Latour
and
others
specifically
sorry
I
have
to
change
the
place
to
put
my
laptop
into
the
charge
again
but
the
thing

is
that
Latour
was
one
of
the
main
people
who
was
complaining
about
this
kind
of
dismissiveness
of
social
aspects
of
scientific
activity
and
I
think
that
people
who
were
working
in
distributed
cognition
were
trying
to
were

actually
explicitly
mentioning
works
of
Latour
and
some
other
social
cognitive
is
to
find
a
foothold
for
bringing
for
stating
cognition
or
knowledge
as
some
sort
of
distributed
social
activity
so
what
I
try
to
do
is
to

develop
some
sorts
of
collective
distributed
account
of
scientific
cognition
it
has
been
something
that
people
such
as
Ronald
Geary
have
been
doing
before
and
Nancy
Narcissian
and
some
other
people
are
doing
these
days
but
what
I

was
trying
to
do
is
to
try
to
reinforce
and
reconstruct
these
very
interesting
insights
by
drawing
on
recent
development
in
free
energy
principle
so
some
sort
of
free
energy
principle
account
of
social
communication
I'm
building
up on

that
to
reconstruct
classical
accounts
of
distributed
scientific
cognition
which
as well
has
its roots
in
works
of
Latour
and
some
other
social
construct
so
you
were
very
sharp
to
spot
the
relation
cool
nice
connection
there
looking
forward
to that

like
actor
network
theory
and
distributed
systems
kind
of
makes
sense
well
I think
if anyone
else
wants to
give
a
final
thought
or
comment
anyone
other
than
the
author
then
we'll
provide
the
last
word
to
the
author
so
blue

and
then
either
Dave
or
Dean
this
was a
lot
of
like
fun
thought
experiments
and
just
thanks
for
your
time
and
as
any
good
paper
opens
more
questions
than
provides
answers
I
think
so
I'm
curious
to
see

what
your
next
work
is
and
how
it
all
unfolds
in
the
future
thank
you
very
much
for
your
contribution
as
well
Dean
and then
Dave
if you
would
like
as
I
mentioned
earlier
I
hope
that
if
in
a

couple
months
time
and
there's
a
slot
where
you
have
availability
I
would
like
to
look
at
that
this
work
that
you
done
around
abduction
and
the
idea
of
active
inference
as
well
because
I
think
in
terms

of
trying
to
be
able
to
cast
a
wider
net
and
get
a
better
sense
of
how
we
model
and
look
at
things
through
through
time
frames
both
before
during
and
after
I
think
that
would
be
something

that
I'd
like
to
take
up
because
going
through
this
paper
certainly
brought
a lot
of
things
to
the
forum
was
really
really
helpful
so
thank
you
very
much
thank
you
for
showing
up
today
it
was
really
appreciated

no
thank
you
very
much
for
your
interest
so
I
learned
quite
a
lot
too
both
from
participating
today
and
from
watching
the
video
the
previous
video
of
the
discussion
great
well
yep
as
always
super
fun
to

read
and
to
interact
with
you
to
take
it
from
a
epistemic
stigmergy
interaction
with
your
asynchronous
artifact
into
something
that is
spatially
remote
but
still
interactive
hashtag
effective
connectivity
so
just
keep in
contact
with us
and
anytime
you
would

like
to
discuss
more
or
do
anything
else
with
the
lab
just
be
in
touch
and
anyone
else
as
well
so
great
times
thank
you
everybody
thank
you
very
much
and
sorry
for
being
a bit
dizzy
after
teaching

and
this
end
of
the
semester
hassle
so
thank
you
very
much
take
care
excellent
see
you
all
later